Year 7 Mathematics

Unit 2: Number Skills

2.5 More Powers and Roots

More Powers and Roots

STARTER · 5 MIN

Date: _____ **Title: More Powers and Roots**

LEARNING OBJECTIVES

- To carry out calculations involving squares, cubes, square roots and cube roots

STARTER *Answer in your book. No calculators.*

1. Work out $2 \times 2 \times 2$

2. Work out 5^2

3. Work out $\sqrt{36}$

4. Work out $3 \times 3 \times 3$

Starter — Answers

STARTER

Tick or correct in green pen

1. **8**

2. **25**

3. **6**

4. **27**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify squares, cubes, square roots and cube roots	☐
SILVER	I can calculate squares, cubes, square roots and cube roots	☐
GOLD	I can apply my knowledge to solve calculations and problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — More Powers and Roots

INTRODUCTION · 3 MIN

A cube number is a number multiplied by itself three times.

Cube number

A number times itself, three times.

$3^3 = 3 \times 3 \times 3 = 27$

Cube root

The inverse of cubing.

$\sqrt[3]{27} = 3$ because $3^3 = 27$

Volume

Volume of a cube = side^3.

Side 4 cm: volume = 64 cm^3

Order

Powers and roots first, then × ÷, then + −.

$3 \times 4^2 = 3 \times 16 = 48$

Examples 1 and 2 — Cube Numbers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $3^3 = 3 \times 3 \times 3$ Cubes: 1, 8, 27, 64, 125

Example 1

Which of these are cube numbers?

8 9 16 27 64

Example 2

Write $5 \times 5 \times 5$ using index notation. Work it out.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Cubes: 1, 8, 27, 64, 125, ...

$2^3 = 8$, $3^3 = 27$, $4^3 = 64$

Cube numbers = 8, 27, 64

Tip: 9 and 16 are square numbers.

Example 2

Three 5s: 5 cubed

$5 \times 5 \times 5 = 5^3$

$= 125$

Tip: 5^3 is not 5×3.

Task 1

PRACTICE · 9 MIN

BRONZE *Work out each cube*

Hint $3^3 = 3 \times 3 \times 3$ Cubes: 1, 8, 27, 64, 125

1. 2^3

2. 4^3

3. 10^3

4. Write $6 \times 6 \times 6$ in index notation

5. Which are cubes?
   1 10 100 1000

★ Which numbers under 100 are both square and cube?

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. 8
2. 64
3. 1000
4. $6^3 = 216$
5. 1 and 1000
★ 1 and 64

Examples 3 and 4 — Cube Roots

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Cube root: undo cubing
$4^3 = 64$, so the cube root of 64 is 4

Example 3

Work out

(a) $\sqrt[3]{64}$

(b) $\sqrt[3]{125}$

Example 4

Work out

$2^3 + \sqrt[3]{8}$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Which number cubed gives it?

(a) $\sqrt[3]{64} = 4$

(b) $\sqrt[3]{125} = 5$

Tip: $4^3 = 64$ and $5^3 = 125$.

Example 4

Powers and roots first

$2^3 = 8$

$\sqrt[3]{8} = 2$

Total $= 8 + 2$

$= 10$

Tip: cube root and cube undo each other.

Task 2

PRACTICE · 9 MIN

SILVER

Work out each answer

Hint Cube root: undo cubing $4^3 = 64$, so the cube root of 64 is 4

1 $\sqrt[3]{8}$

2 $\sqrt[3]{27}$

3 $\sqrt[3]{1000}$

4 $3^3 - 2^3$

5 $\sqrt{49} + \sqrt[3]{64}$

★ $2^3 \times \sqrt{9}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 2
2. 3
3. 10
4. 27 − 8 = 19
5. 7 + 4 = 11

★ 8 × 3 = 24

Examples 5 and 6 — Calculations with Powers and Roots

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Powers and roots first Then × ÷, then + −

Example 5

(2 marks)

Work out

$3 \times 4^2 - \sqrt{81}$

Example 6

(2 marks)

A cube has a volume of 64 cm³.

Work out its side length.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Powers and roots first

$4^2 = 16$

$\sqrt{81} = 9$

$3 \times 16 - 9 = 48 - 9$

$= \mathbf{39}$

Tip: then × before −.

Example 6

Side = cube root of the volume

$\text{Side} = \sqrt[3]{64}$

$= \mathbf{4\ cm}$

Tip: check 4 × 4 × 4 = 64.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: show all your working*

Hint Powers and roots first Then × ÷, then + −

1 2×5^2

2 $5^3 - 10^2$

3 $\sqrt{64} + \sqrt[3]{64}$

4 A cube has volume 1000 cm³.
Find its side length.

5 Square: area 196 cm², side a.
Cube: volume 125 cm³, side b.
Work out $a - b$.

★ Use question 5.
Work out $a^2 - b^3$.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $2 \times 25 = 50$

2. $125 - 100 = 25$

3. $8 + 4 = 12$

4. 10 cm

5. $14 - 5 = 9$

★ $196 - 125 = 71$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Work out 4^3

SILVER Work out $\sqrt[3]{27}$

GOLD Presents: a power of 3 and a power of 2, one apart. How many?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **64**

SILVER **3**

GOLD **Salman 9 (3^2), Rakan 8 (2^3)**

Exam Practice — Cubes

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $3^3 = 3 \times 3 \times 3$ Cubes: 1, 8, 27, 64, 125

Question 1

(2 marks)

Work out each cube.

(a) 3^3 (1)

.....

(b) 10^3 (1)

.....

Question 2

(2 marks)

Here is a list of numbers.

4 8 16 27 36

(a) Write down the cube numbers. (1)

.....

(b) Write down the square numbers. (1)

.....

Exam Practice — Cube Roots

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Cube root: undo cubing
$4^3 = 64$, so the cube root of 64 is 4

Question 3

(2 marks)

Work out each cube root.

(a) $\sqrt[3]{64}$ (1)

.....

(b) $\sqrt[3]{1000}$ (1)

.....

Question 4

(2 marks)

Work out

$\sqrt{81} + 2^3$

.....

Exam Practice — Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Powers and roots first Then × ÷, then + −

Question 5

(2 marks)

Work out the value of

$2 \times 5^2 - \sqrt{64}$

.....

Question 6

(3 marks)

A square has an area of 144 cm^2.
A cube has a volume of 27 cm^3.
The side of the square is a cm.
The side of the cube is b cm.

Work out $a + b$. Show your working.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 27 B1

(b) 1000 B1

Tip: 3 × 3 × 3.

Question 2

(a) 8, 27 B1

(b) 4, 16, 36 B1

Tip: $8 = 2^3$, $27 = 3^3$.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 4 — B1

(b) 10 — B1

Tip: $4^3 = 64$.

Question 4

9 + 8 = 17 — M1 A1

Tip: work out each part first.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$2 \times 25 - 8 = 50 - 8$ **M1**

$= \mathbf{42}$ **A1**

Tip: powers and roots first.

Question 6

$a = \sqrt{144}$

$= 12$ **M1**

$b = \sqrt[3]{27}$

$= 3$ **M1**

$\mathbf{a + b = 15}$ **A1**

Tip: undo the square and the cube.